

Juniorprogrammierer.de

Java 3: Loops & Conditions

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Agenda

- List
- Conditions
- Loop I + II
- Exercise

List

- It is a list of variables
- While in Java an array needs to be for one type, the type in lists can be mixed up
- It allows duplicate members
- Each list starts with index “0” ([0]) and counts up
- It allows negative indexing [-1]

```
1 list1 = ["apple", "banana", "cherry"]
2 list2 = [1, 5, 7, 9, 3]
3 list3 = [True, False, False]
4 list4 = ["abc", 34, True, 40, "male"]
5
6 print(list1[0])
7 print(list1[2])
8 #print(list1[3])
9 print(list4)
10
11 print(list2[-1])
12
13 #length of list
14 len(list1)
15
16 #add a value
17 list1.append("orange")
18
19 #remove a value
20 list2.remove(3)
21
22 #Insert a value
23 list3.insert(1,True)
```

Conditions

- if condition:
 - Indentation xxx
- elif condition:
 - Indentation xxx
- else:
 - Indentation xxx
- match expression:
 - case x:
 - code block
 - case y:
 - code block

```
a = 33
b = 200

if b > a:
    print("b is greater than a")
elif a == b:
    print("a and b are equal")
elif a != b:
    print("xxx")
elif a < b:
    print("sss")
elif a <= b:
    print("eeee")
elif a >= b:
    print("fff")
else:
    print("a is greater than b")
```

```
day = 4
match day:
    case 1:
        print("Monday")
    case 2:
        print("Tuesday")
```

Loops I

for-Loop :

- When you know exactly how often you want to execute something use it with range function
- When you want to loop throw a list use for each loop
- We also got “break” and “continue” cmds

```
62 #prints from 0 to 5
63 for x in range(6):
64     print(x)
65
66 #Starts at 2 and < 30 and inkrements in 3 s1
67 for x in range(2, 30, 3):
68     print(x)
69
70 #For each loop throw a list
71 for x in list1:
72     print(x)
73
74 #example for nested loops
75 adj = ["red", "big", "tasty"]
76 fruits = ["apple", "banana", "cherry"]
77
78 for x in adj:
79     for y in fruits:
80         print(x, y)
81
82 #break
83 fruits = ["apple", "banana", "cherry"]
84 for x in fruits:
85     print(x)
86     if x == "banana":
87         break
88
89 #break
90 for x in fruits:
91     if x == "banana":
92         continue
93     print(x)
94
95 #Use else to specify when loop is finished
96 for x in fruits:
97     print(x)
98 else:
99     print("Finally finished!")
```

Loops II

While-Loop: `i = 1` while condition: `i += 1`

- Is executing as long as condition is true

```
101 #While loop example
102 i = 1
103 while i < 6:
104     print(i)
105     i += 1
106 else:
107     print("i is no longer less than 6")
```

Exercise factorial (Deutsch: Fakultät)

Create an fakultaet.py

- The factorial (Fakultät) of a positive integer n , denoted as $n!$, is the product of all positive integers less than or equal to n
 - $3! = 3 * 2 * 1 = 6$ oder
 - $5! = 5 * 4 * 3 * 2 * 1 = 120$
- Your program has a variable “factorial”. Based on this variable you calculate the factorial and print it out
- Create for every loop a solution!

Exercise leap year (Deutsch: Schaltjahr)

Create an schaltjahr.py.

- A leap year is a year that includes an extra day, February 29th, to keep the calendar year synchronized with the Earth's orbit around the Sun. The rule for determining a leap year is as follows:
 - **Divisibility by 4:** A year is a leap year if it is divisible by 4.
 - **Exception for centuries:** Years that are divisible by 100 are **not** leap years **unless** they are also divisible by 400.
- Your program should have a variable where you enter the year in question. As a result it is printed if the corresponding year is a leap year or not
- Test your code:
 - Das Jahr 2000 ist ein Schaltjahr, da es durch 400 teilbar ist.
 - Das Jahr 1900 ist kein Schaltjahr, da es zwar durch 100, aber nicht durch 400 teilbar ist.
 - Das Jahr 2024 wird ein Schaltjahr sein, da es durch 4 teilbar ist und die anderen Regeln nicht zutreffen.